

Tiny URL

Functional and Non Functional Requirement

Shorten URL: Convert a long URL into a short one.

Functional Requirement

Redirect: When a short URL is accessed, redirect to the original long URL.

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Custom alias: Allow users to specify their own short key.

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Analytics (optional): Track clicks, location, device, etc.

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Expiration (optional): Support expiry dates for short URLs.

Scalability: Handle billions of URLs and millions of requests/day.

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Non-Functional Requirement

User accounts (optional): Allow login for managing URLs.

High availability: Service should be up 24/7.

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Low latency: Redirects must be near-instant.

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Consistency: Short key must always map to the correct long URL.

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Fault tolerance: Survive server crashes with replication.

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Back of the envelop Estimation

Security: Prevent malicious URLs, enforce rate limits.

1. Core Assumptions & Scale

To make the math simple, we use standard system design assumptions:

- **New URLs written (Writes):** 100 million per month
- **Read-to-Write Ratio:** 100:1 (Read-heavy system)
- **URL Expiration / Retention:** 5 years
- **Average URL size:** 500 bytes

Final Estimation Summary

Category	Metric	Calculated Result
Traffic	Write Throughput	40 RPS
	Read Throughput	4,000 RPS (Peak: 8,000 RPS)
Storage (6 Months)	Active Records	600 Million URLs
	Total Primary DB Storage	300 GB
Bandwidth	Incoming Data (Writes)	20 KB/s

	Outgoing Data (Reads)	2 MB/s
Memory	Redis Cache RAM (20% rule)	35 GB

Database Schema

Column Name	Data Type	Constraints / Notes
id	BIGINT	Primary Key, Auto Increment
short_key	VARCHAR(10)	Unique, Not Null (Base62 encoded key or custom)
long_url	TEXT	Not Null (stores original long URL)
created_at	TIMESTAMP	Default = CURRENT_TIMESTAMP
expires_at	TIMESTAMP	Nullable (expiry date for short URL)
user_id	BIGINT	Nullable, Foreign Key (if user accounts exist)
click_count	BIGINT	Default = 0 (tracks number of redirects)
last_accessed	TIMESTAMP	Nullable (last time the short URL was used)
is_active	BOOLEAN	Default = TRUE (soft delete / disable option)

API Methods & Endpoints

1. Create Short URL

- POST /api/shorten
- Request:

```
{
  "long_url": "https://example.com/product?id=123",
  "custom_key": "myalias",
  "expires_at": "2026-12-31"
}
```

- Response:

```
{
  "short_url": "https://tinyurl.com/myalias"
}
```

2. Redirect

- GET /{short_key}
- Server looks up short_key → returns HTTP 302 redirect to long_url.

3. Analytics

- GET /api/stats/{short_key}
- Response:

```
{
  "short_key": "abc123",
  "long_url": "https://example.com/product?id=123",
  "click_count": 542,
  "created_at": "2026-07-31"
}
```

4. Delete URL (optional)

- DELETE /api/{short_key}

System Architecture

